

- (a)** Aphids are small insects which feed directly on phloem sap.

The salivary glands of aphids have secretory cells that make and release a variety of proteins that assist in feeding.

Fig. 1.1 is a transmission electron micrograph of a small area of a salivary gland cell of an aphid.



Fig. 1.1

Describe the role of Golgi bodies in secretory cells, such as the salivary gland cells of aphids.

..... [3]

(b) (i) Explain why secretory cells have large numbers of mitochondria.

.....

.....

.....

..... [2]

(ii) Mitochondria are partly controlled by the nucleus, but can also function independently.

Suggest the features of mitochondria that allow them to function independently of the nucleus.

.....

.....

.....

.....

.....

..... [2]

(c) Aphids are important vectors of plant viral diseases.

(i) Describe the structure of a typical virus.

.....

.....

.....

.....

.....

..... [3]

(ii) Suggest how viruses are able to pass from one plant cell to the next without crossing membranes.

.....

.....

..... [1]

[Total: 11]